Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 9

Conjunctive Normal Form, Satisfiability, and SAT

Covers *MCS* Sections 3.3.2, 3.4.1, and 3.5

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Everything flips

The two recipes side by side

	Built from	Shape	Variable that is T in that row
DNF	**true** rows	OR of ANDs	appears plain
CNF	**false** rows	AND of ORs	appears *negated*

Count the F's before writing a CNF. That is how many clauses you will have.

Three words, and how they relate to Monday

Valid = true in every row = Chartrand's *tautology*.
Satisfiable = true in at least one row.
Unsatisfiable = true in no row = Chartrand's *contradiction*.
Every valid formula is satisfiable; the converse fails. And S is satisfiable if and only if $\sim S$ is not valid.

Three mistakes to expect

1. **Using the DNF sign convention.** In a CNF clause, a variable that is T in the row appears *negated*.
2. **Confusing valid with satisfiable.** "There is an assignment making it true" is a far weaker claim than "every assignment does."
3. **Claiming unsatisfiable after testing a few assignments.** That is the sampling error from Week 1, wearing a new hat.

Part II: Practice Problems

Section A — Reading CNF off a table

1. Build the truth table for $P \oplus Q$, then write its CNF.

P	Q	$P \oplus Q$	Clause
T	T		
T	F		
F	T		
F	F		

2. Write the CNF of each.

(a) $P \wedge Q$

(b) $P \vee Q$

(c) $\sim P$ (over the variables P and Q)

Section B — Validity and satisfiability

3. Classify each as valid, satisfiable but not valid, or unsatisfiable. Justify briefly.

(a) $(P \Rightarrow Q) \vee (Q \Rightarrow P)$

(b) $P \wedge (\sim P)$

(c) $(P \vee Q) \wedge (\sim P)$

(d) $\big(P \wedge (P \Rightarrow Q)\big) \Rightarrow Q$

4. Determine whether each CNF formula is satisfiable. Give a satisfying assignment or argue that none exists.

(a) $(P \vee Q) \wedge (\sim P \vee Q) \wedge (\sim Q)$

(b) $(P \vee Q) \wedge (\sim P \vee R) \wedge (\sim Q \vee \sim R)$

Section C — Specifications

5. A build system must satisfy all three requirements at once:

(1) If the tests pass, the artifact is published.

(2) The artifact is not published, or the version is tagged.

(3) The tests pass, and the version is not tagged.

Let T, A, and V stand for the three claims. Write each requirement as a clause, then determine whether all three can hold together.

Section D — Writing

6. In five to eight complete sentences, explain why building a truth table is an impractical way to solve SAT for large formulas, and why this matters beyond this course. Use a specific number of variables in your explanation.

Section E — Challenge (optional)

7. Show how to obtain a CNF for a formula from the DNF of its *negation*.